

SMEDEREVSKA PALANKA, Serbia

WMO: 132790

Lat: **44.367N**Lon: **20.950E**Elev: **400**StdP: **14.48**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	17.7	7.7	8.3	13.6	12.8	10.7	19.0	18.2	35.2	16.3	36.3	3.5	320	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	94.0	70.1	90.7	69.8	87.5	69.0	73.1	87.4	71.6	85.6	70.2	83.9	5.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.9	107.6	78.9	67.1	101.2	76.9	65.7	96.2	75.3	36.9	87.4	35.6	86.0	34.3	83.9	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	13.9	11.9	2.5	99.1	8.8	4.6	-3.8	102.3	-9.0	105.0	-13.9	107.6	-20.3	110.9		
(5)			1.9	75.6	8.4	1.8	-4.1	76.9	-9.0	78.0	-13.7	79.0	-19.8	80.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	33.9	37.6	45.1	54.3	62.4	68.9	72.4	72.3	63.7	54.1	45.7	36.2		
	DBStd	15.54	9.19	9.59	8.48	7.19	6.46	6.32	5.60	5.61	6.82	7.94	8.56	8.65		
	HDD50	1761	501	357	196	40	1	0	0	0	51	180	435			
	HDD65	4772	964	768	616	326	127	33	8	9	105	344	578	894		
	CDD50	3209	2	9	45	168	387	566	693	692	410	178	53	6		
	CDD65	744	0	0	0	4	47	149	236	236	65	7	0	0		
	CDH74	8391	0	0	4	84	563	1666	2669	2626	700	79	0	0		
	CDH80	3352	0	0	0	13	117	615	1176	1195	228	8	0	0		
(14) Wind	WSAvg	4.9	5.1	5.5	5.9	5.5	4.7	4.5	4.4	4.2	4.4	4.5	5.2	5.1		
(15) Precipitation	PrecAvg	26.1	1.7	1.6	1.8	2.1	2.6	3.0	2.6	2.1	2.4	2.2	1.9	2.1		
	PrecMax	40.7	3.4	3.0	4.6	5.1	9.4	6.0	9.8	5.2	6.2	4.6	4.5	5.1		
	PrecMin	14.9	0.4	0.4	0.2	0.1	0.7	0.9	0.3	0.0	0.5	0.0	0.2	0.3		
	PrecStd	5.5	0.8	0.8	1.2	1.1	1.7	1.3	2.0	1.4	1.4	1.3	0.9	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.3	66.2	72.9	81.5	86.9	93.6	98.6	99.1	92.4	80.7	70.4	60.0		
		MCWB	49.8	52.2	55.0	62.1	66.1	71.0	70.0	70.5	68.2	63.5	58.2	51.8		
	2%	DB	53.5	60.6	68.3	75.7	83.2	90.0	93.7	94.6	86.1	75.9	66.5	55.7		
		MCWB	47.0	49.9	53.1	58.7	65.2	70.9	70.3	69.7	66.1	62.1	56.2	49.4		
	5%	DB	50.0	55.8	63.9	71.8	80.3	86.8	90.4	90.6	81.5	71.7	62.4	51.9		
		MCWB	45.1	47.6	51.7	56.8	64.2	69.7	70.0	69.3	64.7	60.4	53.9	46.7		
	10%	DB	46.5	51.6	59.5	68.2	77.0	83.5	86.7	86.9	77.4	67.8	58.7	48.3		
		MCWB	42.8	45.4	49.3	55.3	63.0	68.3	69.2	68.6	63.4	58.5	51.9	44.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.2	53.3	56.8	63.3	68.6	75.7	74.5	75.5	70.1	65.9	59.3	53.2		
		MCDB	57.8	63.7	70.2	78.6	82.0	88.0	87.8	88.5	86.1	77.7	68.7	58.0		
	2%	WB	47.4	50.4	54.3	60.3	66.6	72.6	72.8	72.8	67.5	63.1	56.6	49.8		
		MCDB	52.6	59.3	65.8	73.1	80.0	85.8	87.3	87.3	81.7	73.1	65.1	55.1		
	5%	WB	45.3	48.1	52.2	58.2	65.2	70.8	71.4	71.1	65.9	61.1	54.6	47.1		
		MCDB	49.5	55.1	62.6	69.3	78.0	83.5	85.6	85.6	78.6	69.8	61.4	51.7		
	10%	WB	42.9	45.8	50.0	56.1	63.6	69.1	70.0	69.6	64.3	59.2	52.4	44.6		
		MCDB	46.5	51.4	58.6	65.9	75.0	81.3	83.7	83.6	75.1	66.9	58.2	47.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	14.6	18.0	19.9	20.7	21.0	22.7	23.5	20.6	18.8	14.8	11.2		
		MCDBR	17.5	23.0	26.4	27.8	27.3	27.6	29.8	31.6	28.2	25.1	20.6	16.8		
		MCWBR	12.0	14.0	14.1	13.6	11.8	10.8	10.1	10.5	11.1	13.0	11.9	11.5		
	5% WB	MCDBR	16.6	21.2	23.6	24.0	24.4	24.4	24.5	25.2	24.1	23.0	18.6	15.8		
		MCWBR	11.7	13.2	12.9	12.3	11.3	10.3	9.4	9.6	10.2	12.4	10.9	11.1		
(40) Clear-Sky Solar Irradiance	taub		0.344	0.371	0.405	0.463	0.451	0.445	0.448	0.460	0.436	0.398	0.360	0.341		
	taud		2.322	2.225	2.198	2.053	2.148	2.207	2.200	2.162	2.197	2.298	2.354	2.347		
	Ebn at Noon		240	253	260	254	261	263	261	252	247	241	232	228		
	Edh at Noon		26	34	40	50	47	45	44	44	40	32	25	23		
(44) All-Sky Solar Radiation	RadAvg		403	664	1044	1423	1727	1932	1966	1751	1228	840	516	345		
	RadStd		57	86	128	153	176	192	154	168	129	102	51	50		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.21	N/A	N/A	N/A	+0.74	+1.10	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)		+1.21	N/A	N/A	N/A	+0.67	N/A	N/A	-284	+185	N/A

Nomenclature: See separate page